



实验 两个网关之间通过手工 配置 IPSec VPN 隧道

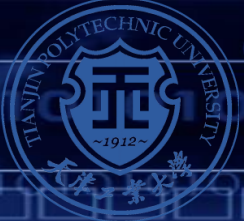
(采用预共享密钥认证)



目 标

❖ 学完本章内容后，同学们将能够：

- 了解通过 IKE 方式协商 IPSec VPN 隧道VPN 技术原理
- 掌握防火墙中两个网关之间通过 IKE 方式协商 IPSec VPN 隧道（采用预共享密钥认证）VPN 配置流程



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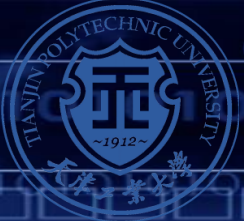
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IKE 方式协商 IPSec VPN 隧道配置



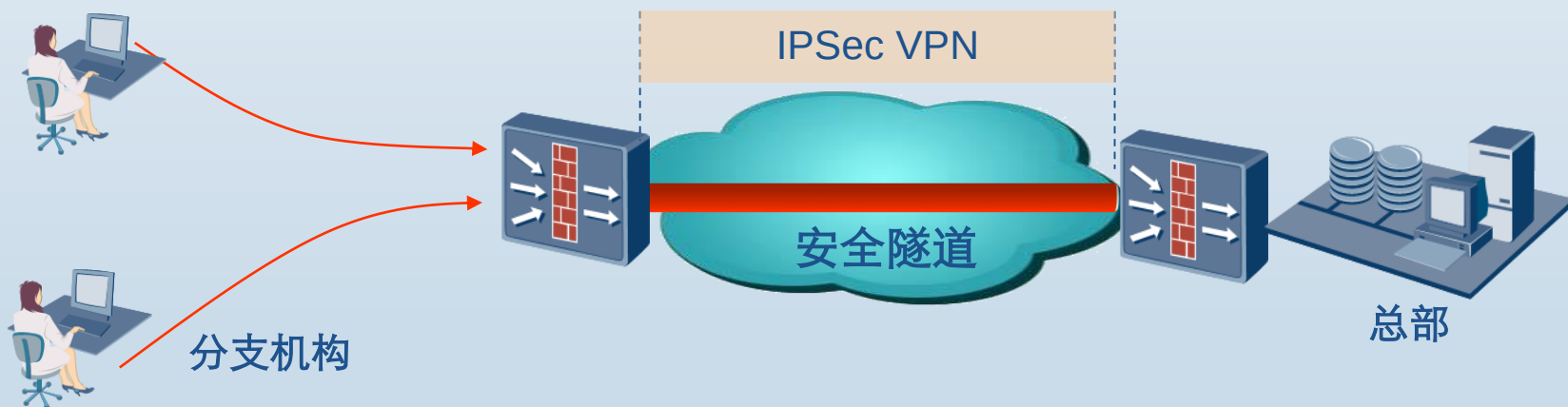
1

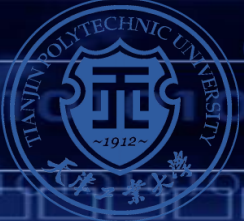
IPSec VPN 隧道基本原理



IPSec简介

- ❖ IPSec (IP Security) 协议族是IETF制定的一系列安全协议，它为端到端IP报文交互提供了基于密码学的、可互操作的、高质量的安全保护机制。IPSec VPN是利用IPSec隧道建立的网络层VPN。

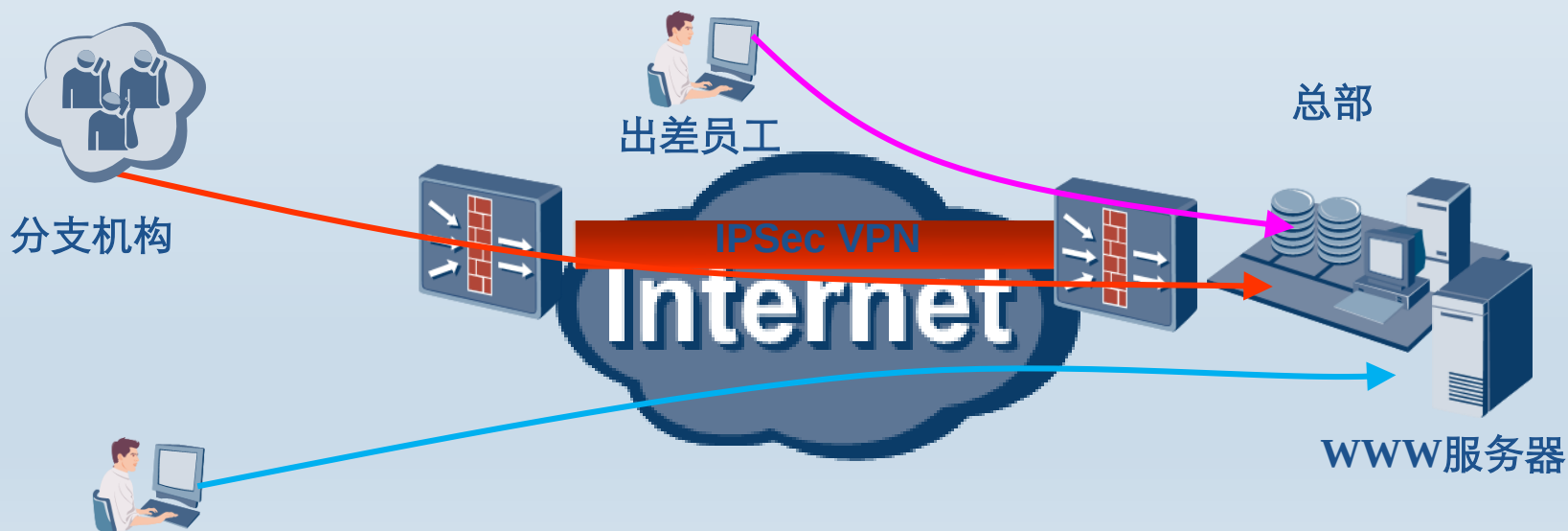


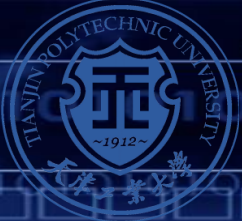


IPSec安全防护场景

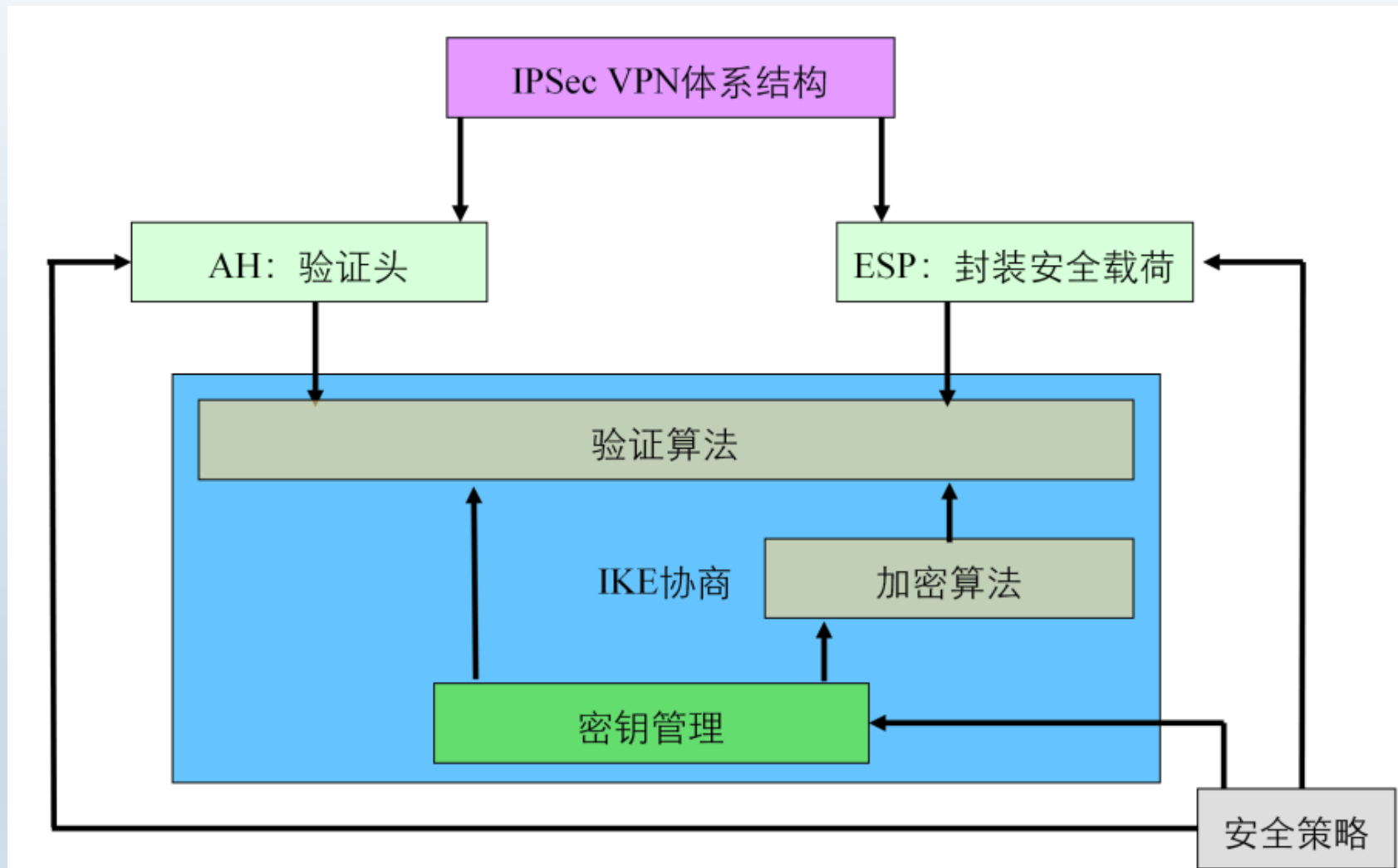
❖ IPSec端到端应用场景

- ❑ 安全网关（如防火墙）之间（典型场景）；
- ❑ 主机与安全网关之间；
- ❑ 主机与主机之间；





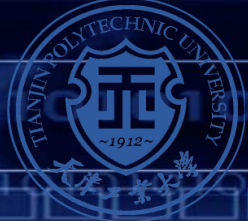
IPsec组成部分





2

实验需求与结构搭建



实验需求

❖ 某公司的总部和分支机构均使用FW作为网关。网络A和网络B分别通过FW_A和FW_B连接到Internet。网络环境描述如下：

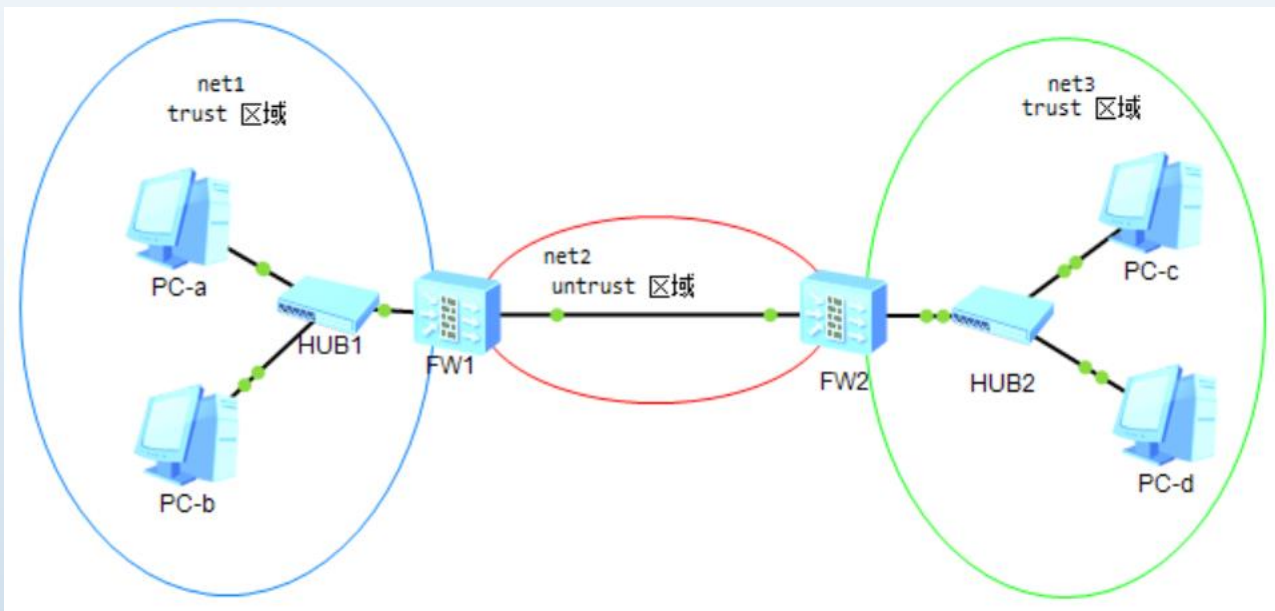
- 网络A属于1.1.1.0/24子网，通过接口GigabitEthernet 1/0/0与FW-A连接。
- 网络B属于3.3.3.0/24子网，通过接口GigabitEthernet 1/0/0与FW-B连接。
- FW-A和FW-B路由可达。

❖ 通过组网实现如下需求：

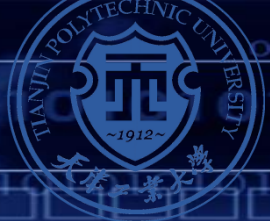
- 在FW-A和FW-B之间建立IPSec隧道，
- 使net1和 net3的用户可通过IPSec隧道互相访问。



设备与拓扑



实验设备：
USG6000: 2台
HUB: 2台
PC: 4台



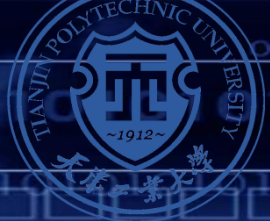
数据规划

网络设备名称	接口	IP地址	所属安全区域	互联对端设备	对端接口
FW1	G1/0/0	2.x.y.1/24	untrust	FW2	G1/0/0
	G1/0/1	1.x.y.254/24	trust	HUB1	e0/0/0
FW2	G1/0/0	2.x.y.2/24	untrust	FW2	G1/0/0
	G1/0/1	3.x.y.254/24	trust	HUB2	e0/0/0
HUB1	e0/0/0	无	无	FW1	G1/0/1
	e0/0/1	无	无	PC-a	
	e0/0/2	无	无	PC-b	
HUB2	e0/0/0	无	无	FW2	
	e0/0/1	无	无	PC-c	
	e0/0/2	无	无	PC-d	

主机名称	IP地址	子网掩码	默认网关
PC-a	1.x.y.1	255.255.255.0	1.x.y.254
PC-b	1.x.y.2	255.255.255.0	1.x.y.254
PC-c	3.x.y.1	255.255.255.0	3.x.y.254
PC-d	3.x.y.2	255.255.255.0	3.x.y.254

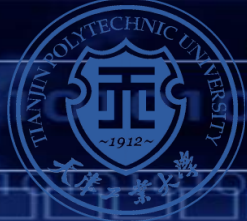
特别注意：IP地址中的x,y的含义

x：学号倒数第三位； y:学号倒数后两位



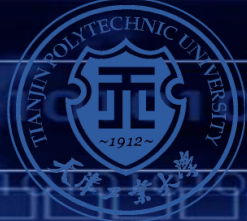
防火墙数据准备

- ❖ 采用的安全协议，以及安全协议使用的加密算法和验证算法
- ❖ 安全协议采用ESP协议
 - ESP协议采用的加密算法为AES-256
 - ESP协议采用的验证算法为SHA2-256。
- ❖ 入方向和出方向的安全参数索引(SPI)。
 - FW1
 - 入方向安全联盟的SPI为12345,
 - 出方向安全联盟的SPI为54321。
 - FW2
 - 入方向安全联盟的SPI为54321
 - 出方向安全联盟的SPI为12345
- ❖ ESP协议的验证密钥。
 - FW1采用的
 - 入方向上ESP协议的验证密钥为abcd
 - 出方向上ESP协议的验证密钥为dcba
 - FW_B采用的
 - 入方向上ESP协议的验证密钥为dcba
 - 出方向上ESP协议的验证密钥为abcd



接口规划

设备名称	项目	数据	说明
防火墙 FW1	配置高级ACL 3000	源网段：1.1.1.0/24 目的网段：3.3.3.0/24	定义被保护的数据流
	配置IPSec安全提议tran1	encapsulation-mode tunnel transform esp esp authentication-algorithm sha2-256 esp encryption-algorithm aes-256	隧道模式 Esp封装 验证算法 sha2-256 加密算法aes-256
	配置名称为map1，序号为10的IPSec策略	ipsec policy map1 10 manual security acl 3000 proposal tran1 tunnel remote 2.2.2.2 tunnel local 2.2.2.1 sa spi outbound esp 12345 sa spi inbound esp 54321 sa string-key outbound esp abcd sa string-key inbound esp dcba	定义名为map1序号为10的IPSec策略 过滤流量acl3000 安全提议 tran1 隧道对端接口地址2.2.2.2 隧道本地接口地址2.2.2.1 本地去往对端安全联盟标识12345 对端到达本地安全联盟标识54321 本地去往对端安全验证密钥abcd 对端到达本地安全验证密钥dcba
	应用IPSec安全策略	interface GigabitEthernet 1/0/0 ipsec policy map1	在隧道接口上应用IPSec安全策略
	路由	目的网络：3.3.3.0/24 下一跳地址:2.2.2.2	配置静态路由，通往分支网络B的下一跳设备的IP地址为2.2.2.2



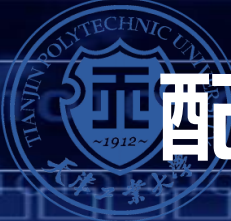
接口规划

设备名称	项目	数据	说明
防火墙 FW2	配置高级ACL 3000	源网段: 1.1.1.0/24 目的网段: 3.3.3.0/24	定义被保护的数据流
	配置IPSec安全提议tran1	encapsulation-mode tunnel transform esp esp authentication-algorithm sha2-256 esp encryption-algorithm aes-256	隧道模式 Esp封装 验证算法 sha2-256 加密算法aes-256
	配置名称为map1, 序号为10的IPSec策略	ipsec policy map1 10 manual security acl 3000 proposal tran1 tunnel remote 2.2.2.1 tunnel local 2.2.2.2 sa spi outbound esp 54321 sa spi inbound esp12345 sa string-key outbound esp dcba sa string-key inbound esp abcd	定义名为map1序号为10的IPSec策略 过滤流量acl3000 安全提议 tran1 隧道对端接口地址2.2.2.2 隧道本地接口地址2.2.2.1 本地去往对端安全联盟标识54321 对端到达本地安全联盟标识12345 本地去往对端安全验证密钥dcba 对端到达本地安全验证密钥abcd
	应用IPSec安全策略	interface GigabitEthernet 1/0/0 ipsec policy map1	在隧道接口上应用IPSec安全策略
	路由	目的网络: 1.1.1.0/24 下一跳地址:2.2.2.1	配置静态路由, 通往分支网络B的下一跳设备的IP地址为2.2.2.1



3

实验前置配置



配置PC

PC-a

基础配置 命令行 组播 UDP发包工具 串口

主机名: PC-a

MAC 地址: 54-89-98-24-78-2D

IPv4 配置

☒ 静态 ☐ DHCP ☐ 自动获取 DNS 服务器地址

IP 地址: 1 . 1 . 1 . 1

子网掩码: 255 . 255 . 255 . 0

网关: 1 . 1 . 1 . 254

DNS1: 0 . 0 . 0 . 0

DNS2: 0 . 0 . 0 . 0

IPv6 配置

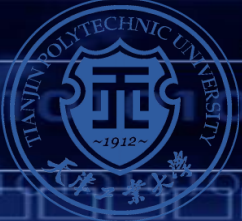
☒ 静态 ☐ DHCPv6

IPv6 地址: ::

前缀长度: 128

IPv6 网关: ::

应用

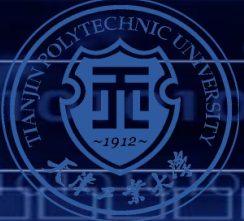


启动设备-首次登录

- ❖ 防火墙首次启动登录需要验证用户身份
- ❖ 采用系统初始默认用户名和密码
- ❖ 首次登录后需要立即修改密码
- ❖ 修改为统一密码，方便日后重复调试

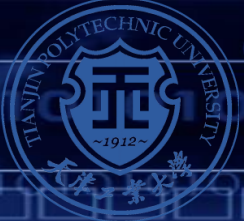
```
FW1
FW1
Username:
Error: The username times out.
Username:admin
Password:
Error: Authentication fail
Username:admin
Password:
The password needs to be changed. Change now? [Y/N]: y
Please enter old password:
Please enter new password:
Please confirm new password:

Info: Your password has been changed. Save the change to survive
*****
*           Copyright (C) 2014-2018 Huawei Technologies Co., Ltd.
*           All rights reserved.
*           Without the owner's prior written consent,
*           no decompiling or reverse-engineering shall be allowed.
*****
<USG6000V1>
```



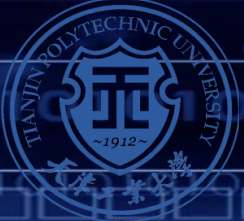
设备首次开启

- ❖ 默认用户名: admin
- ❖ 默认密码: Admin@123
- ❖ 设置新密码: admin@123



华为防火墙命令行模式

- ❖ **sysname**：配置设备名称
- ❖ **Interface g0/0/0**：进入接口配置模式
- ❖ **display**：用于在任何视图下显示相关内容
- ❖ **Ctrl+z** 快捷方式，退出到<> 用户模式
- ❖ **save**：在用户视图模式下完成配置文件的保存
- ❖ **display ip routing-table**：显示路由表
- ❖ **display zone**：显示区域
- ❖ **display firewall session table**：显示防火墙会话表
- ❖ **display security-policy all**（自定义名称）：显示安全策略



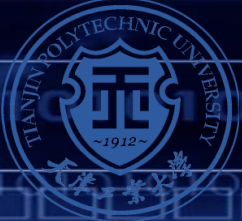
配置防火墙FW1

❖ 修改设备名称

- sysname 学号后三位-1

❖ 配置接口地址

- interface GigabitEthernet1/0/0 (配置接口G1/0/0)
- ip address 2.x.y.1 255.255.255.0(配置IP地址)
- service-manage ping permit (开启ping 服务)
- interface GigabitEthernet1/0/1
- ip address 1.x.y.254 255.255.255.0
- service-manage ping permit



配置防火墙FW1

❖ 配置安全区域

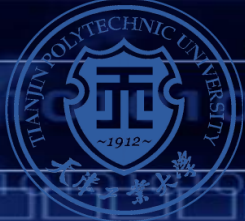
- firewall zone trust
- add interface GigabitEthernet1/0/1
- firewall zone untrust
- add interface GigabitEthernet1/0/0

❖ 配置缺省缺省路由

- ip route-static 0.0.0.0 0.0.0.0 2.x.y.2

❖ 保存配置

- 在普通用户模式下， save



配置防火墙FW1

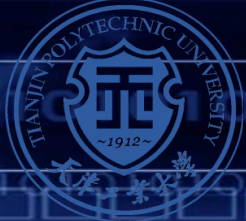
配置安全策略

- 配置Trust域与Untrust域之间的域间安全策略

```
[FW_A] security-policy
[FW_A-policy-security] rule name policy1
[FW_A-policy-security-rule-policy1] source-zone trust
[FW_A-policy-security-rule-policy1] destination-zone untrust
[FW_A-policy-security-rule-policy1] source-address 1.x.y.0 24
[FW_A-policy-security-rule-policy1] destination-address 3.x.y.0 24
[FW_A-policy-security-rule-policy1] action permit
[FW_A-policy-security-rule-policy1] quit
[FW_A-policy-security] rule name policy2
[FW_A-policy-security-rule-policy2] source-zone untrust
[FW_A-policy-security-rule-policy2] destination-zone trust
[FW_A-policy-security-rule-policy2] source-address 3.x.y.0 24
[FW_A-policy-security-rule-policy2] destination-address 1.x.y.0 24
[FW_A-policy-security-rule-policy2] action permit
[FW_A-policy-security-rule-policy2] quit
```

- 配置Untrust域与Local域之间的域间安全策略。

```
[FW_A-policy-security] rule name policy3
[FW_A-policy-security-rule-policy3] source-zone local
[FW_A-policy-security-rule-policy3] destination-zone untrust
[FW_A-policy-security-rule-policy3] source-address 2.x.y.1 32
[FW_A-policy-security-rule-policy3] destination-address 2.x.y.2 32
[FW_A-policy-security-rule-policy3] action permit
[FW_A-policy-security-rule-policy3] quit
[FW_A-policy-security] rule name policy4
[FW_A-policy-security-rule-policy4] source-zone untrust
[FW_A-policy-security-rule-policy4] destination-zone local
[FW_A-policy-security-rule-policy4] source-address 2.x.y.2 32
[FW_A-policy-security-rule-policy4] destination-address 2.x.y.1 32
[FW_A-policy-security-rule-policy4] action permit
[FW_A-policy-security-rule-policy4] quit
[FW_A-policy-security] quit
```



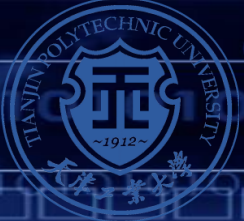
配置防火墙FW2

❖ 修改设备名称

- sysname 学号后三位-2

❖ 配置接口地址

- interface GigabitEthernet1/0/0 (配置接口G1/0/0)
- ip address 2.x.y.2 255.255.255.0(配置IP地址)
- service-manage ping permit (开启ping 服务)
- interface GigabitEthernet1/0/1
- ip address 3.x.y.254 255.255.255.0
- service-manage ping permit



配置防火墙FW2

❖ 配置安全区域

- firewall zone trust
- add interface GigabitEthernet1/0/1
- firewall zone untrust
- add interface GigabitEthernet1/0/0

❖ 配置缺省安全策略

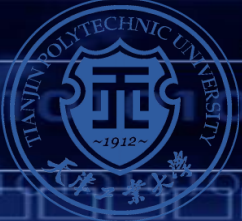
- security-policy
- default action permit

❖ 配置缺省缺省路由

- ip route-static 0.0.0.0 0.0.0.0 2.x.y.1

❖ 保存配置

- 在普通用户模式下， save



PC IP地址配置

PC-a

基础配置 命令行 组播 UDP发包工具 串口

主机名: PC-a

MAC 地址: 54-89-98-24-78-2D

IPv4 配置

☒ 静态 ☐ DHCP ☐ 自动获取 DNS 服务器地址

IP 地址: 1 . 1 . 1 . 1

子网掩码: 255 . 255 . 255 . 0

网关: 1 . 1 . 1 . 254

DNS1: 0 . 0 . 0 . 0

DNS2: 0 . 0 . 0 . 0

IPv6 配置

☒ 静态 ☐ DHCPv6

IPv6 地址: ::

前缀长度: 128

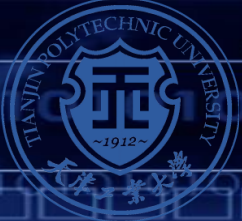
IPv6 网关: ::

应用

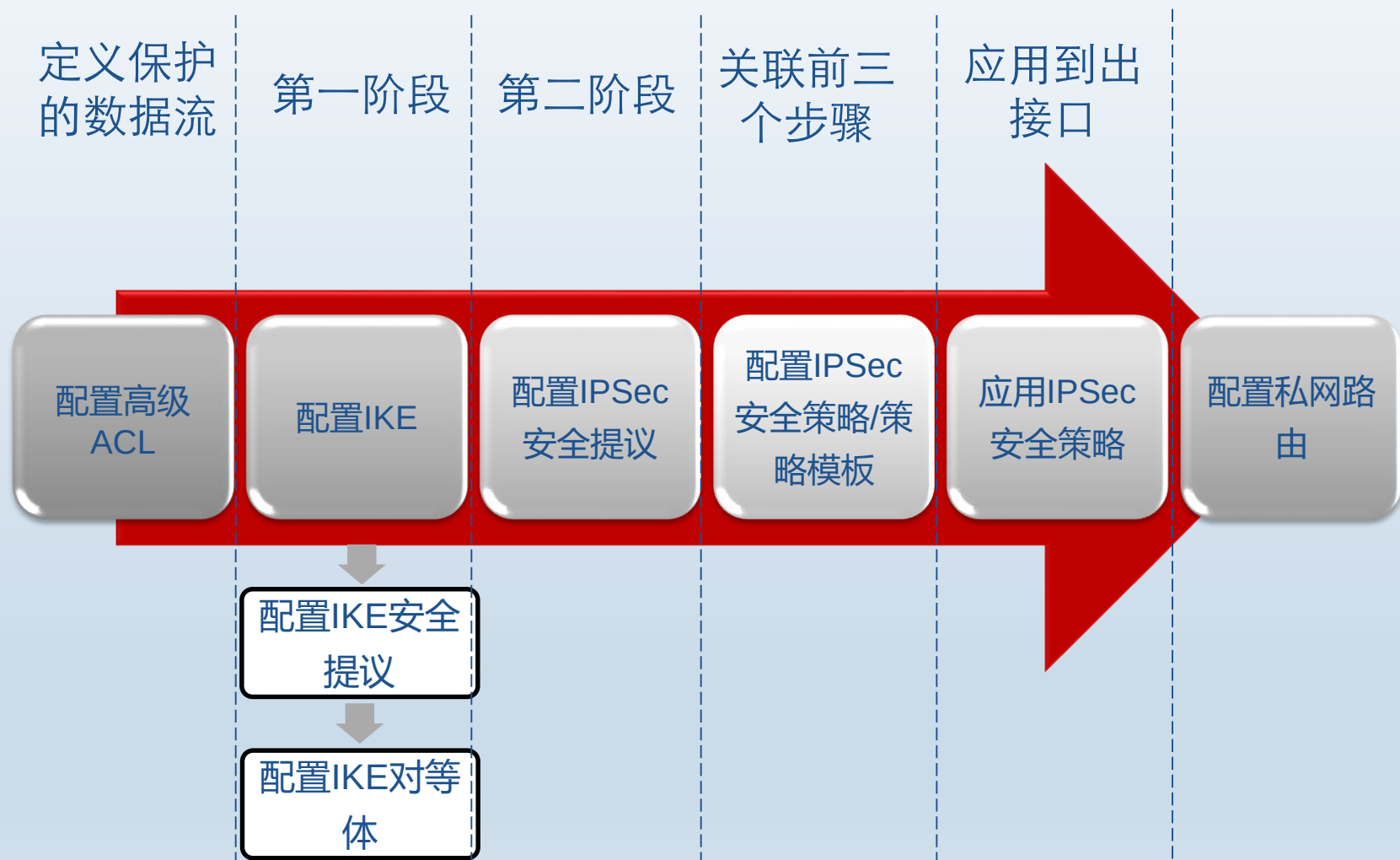


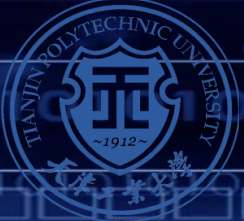
4

IPSec VPN 配置



IPSec VPN配置关键步骤





IPSec VPN配置思路—FW1

- ❖ 定义被保护的数据流。配置高级ACL 3000，允许1.1.1.0/24网段访问3.3.3.0/24网段

```
acl 3000
rule permit ip source 1.1.1.0 0.0.0.255 destination 3.3.3.0 0.0.0.255
quit
```

- ❖ 配置IPSec安全提议tran1

```
ipsec proposal tran1
encapsulation-mode tunnel
transform esp
esp authentication-algorithm sha2-256
esp encryption-algorithm aes-256
quit
```

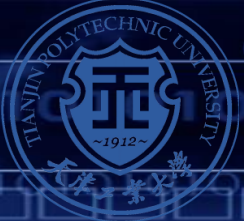
- ❖ 查看安全提议tran1

```
display ipsec proposal
```

```
[test-1]display ipsec proposal
2023-04-24 08:51:16.500
```

```
Number of proposals: 1
```

```
IPSec proposal name: tran1
Encapsulation mode: Tunnel
Transform           : esp-new
ESP protocol        : Authentication SHA2-HMAC-256
                    : Encryption AES-256
```



IPSec VPN配置—FW1

❖ 配置名称为map1，序号为10的IPSec策略

```
ipsec policy map1 10 manual
security acl 3000
proposal tran1
tunnel remote 2.2.2.2
tunnel local 2.2.2.1
sa spi outbound esp 12345
sa spi inbound esp 54321
sa string-key outbound esp abcd
sa string-key inbound esp dcba
quit
```

❖ 显示IPSec策略

```
display ipsec policy
```

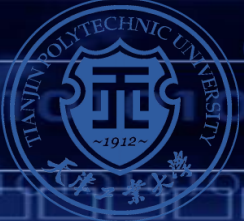
❖ 在接口GigabitEthernet 1/0/0上应用IPSec策略组map1

```
interface GigabitEthernet 1/0/0
ipsec policy map1
quit
```

```
[test-1]display ipsec policy
2023-04-24 08:54:57.420

=====
IPSec policy group: "map1"
Using interface:
=====

Sequence number: 10
Security data flow: 3000/IPv4
Tunnel local address: 2.2.2.1
Tunnel remote address: 2.2.2.2
Proposal name: tran1
Inbound AH setting:
  AH SPI:
  AH string-key:
  AH authentication hex key:
Inbound ESP setting:
  ESP SPI: 54321 (0xd431)
  ESP string-key: %^%#GrvpFfXoFEH`KlG2*QX4!7^.6lUZ^Y&s$^'5b; &@%#%#
  ESP encryption hex key:
  ESP authentication hex key:
Outbound AH setting:
  AH SPI:
  AH string-key:
  AH authentication hex key:
Outbound ESP setting:
  ESP SPI: 12345 (0x3039)
  ESP string-key: %^%#nTl\=#b*XEP6WFGC) IK: |, x (&#4[ ]lAmEdAd/aj-%^%#
  ESP encryption hex key:
  ESP authentication hex key:
Anti-replay: -
Anti-replay window size: -
Fragment before-encryption: Disable
Respond-only: Disable
Policy status : Inactive
Flow-vrf check : Enable
Sa keep-holding-to hard-duration : Disable
```



IPSec VPN配置—FW2

- ❖ 定义被保护的数据流。配置高级ACL 3000，允许3.3.3.0/24网段访问1.1.1.0/24网段

```
acl 3000
rule permit ip source 3.3.3.0 0.0.0.255 destination 1.1.1.0 0.0.0.255
quit
```

- ❖ 配置IPSec安全提议tran1

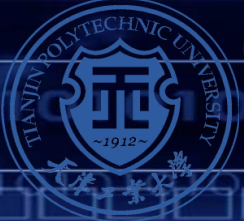
```
ipsec proposal tran1
encapsulation-mode tunnel
transform esp
esp authentication-algorithm sha2-256
esp encryption-algorithm aes-256
Quit
```

- ❖ 查看安全提议tran1

```
disp ipsec proposal name tran1
```

```
[test-2]disp ipsec proposal name tran1
2023-04-24 09:06:16.780
```

```
IPSec proposal name: tran1
Encapsulation mode: Tunnel
Transform           : esp-new
ESP protocol        : Authentication SHA2-HMAC-256
                    : Encryption AES-256
```



IPSec VPN配置—FW2

❖ 配置名称为map1，序号为10的IPSec策略

```
ipsec policy map1 10 manual
security acl 3000
proposal tran1
tunnel remote 2.2.2.1
tunnel local 2.2.2.2
sa spi outbound esp 54321
sa spi inbound esp 12345
sa string-key outbound esp dcba
sa string-key inbound esp abcd
quit
```

❖ 查看IPSec策略

```
display ipsec policy
```

❖ 在接口GigabitEthernet 1/0/0上应用IPSec策略组map1

```
interface GigabitEthernet 1/0/0
ipsec policy map1
quit
```

```
[test-2] display ipsec policy
2023-04-24 09:08:42.050

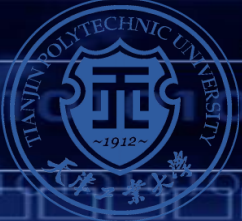
=====
IPSec policy group: "map1"
Using interface:
=====

Sequence number: 10
Security data flow: 3000/Tpvd
Tunnel local address: 2.2.2.2
Tunnel remote address: 2.2.2.1
Proposal name: -
Inbound AH setting:
AH SPI:
AH string-key:
AH authentication hex key:
Inbound ESP setting:
ESP SPI: 12345 (0x3039)
ESP string-key: %*%#-(G4)5Mt%BC)6M+fs.(ytas5S[K2HQY508a7V%*%#
ESP encryption hex key:
ESP authentication hex key:
Outbound AH setting:
AH SPI:
AH string-key:
AH authentication hex key:
Outbound ESP setting:
ESP SPI: 54321 (0xd431)
ESP string-key: %*%#(%>xCp<-B0!qk/@ch@XO[Rb;0xVuV>7`3dCdUALW%*%#
ESP encryption hex key:
ESP authentication hex key:
Anti-replay: -
Anti-replay window size: -
Fragment before-encryption: Disable
Respond-only: Disable
Policy status : Inactive
Flow-vrf check : Enable
Sa keep-holding-to hard-duration : Disable
```



5

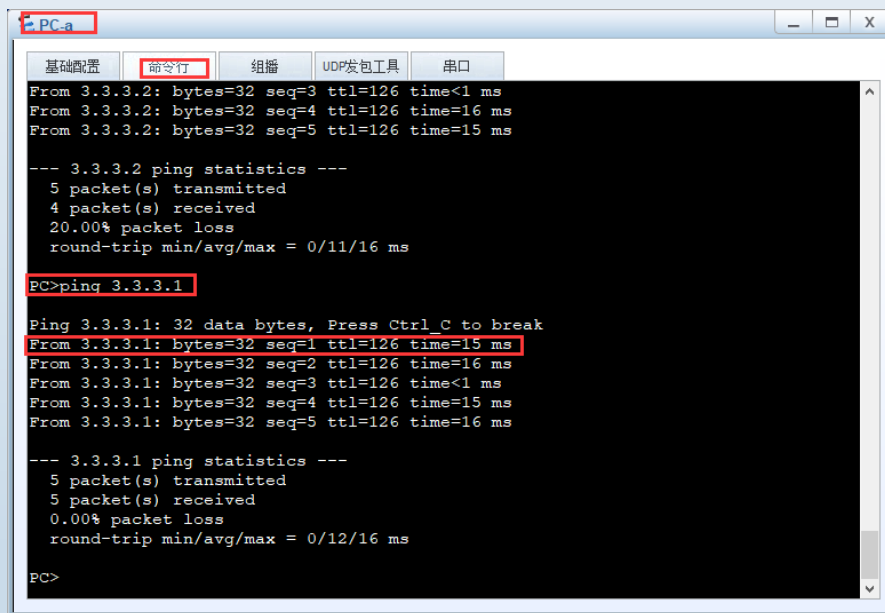
实验结果



PCA->PCc

❖ 配置完成后，在PC-a执行ping命令，ping PC-c。

通畅



```
PC-a
基础配置 命令行 组播 UDP发包工具 串口
From 3.3.3.2: bytes=32 seq=3 ttl=126 time<1 ms
From 3.3.3.2: bytes=32 seq=4 ttl=126 time=16 ms
From 3.3.3.2: bytes=32 seq=5 ttl=126 time=15 ms

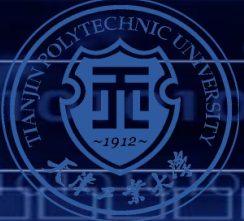
--- 3.3.3.2 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/11/16 ms

PC>ping 3.3.3.1

Ping 3.3.3.1: 32 data bytes, Press Ctrl_C to break
From 3.3.3.1: bytes=32 seq=1 ttl=126 time=15 ms
From 3.3.3.1: bytes=32 seq=2 ttl=126 time=16 ms
From 3.3.3.1: bytes=32 seq=3 ttl=126 time<1 ms
From 3.3.3.1: bytes=32 seq=4 ttl=126 time=15 ms
From 3.3.3.1: bytes=32 seq=5 ttl=126 time=16 ms

--- 3.3.3.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 0/12/16 ms

PC>
```



[FW-B]display ipsec sa

IPSec 连接成功

```
[test-1]disp ipsec sa
2023-04-24 09:27:24.000

ipsec sa information:

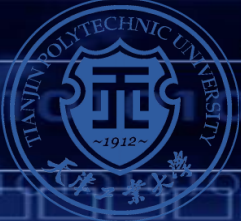
=====
Interface: GigabitEthernet1/0/0
=====

-----
IPSec policy name: "map1"
Sequence number   : 10
Acl group         : 3000
Acl rule          : -
Mode              : Manual
-----

Encapsulation mode: Tunnel
Tunnel local       : 2.2.2.1:0
Tunnel remote      : 2.2.2.2:0

[Outbound ESP SAs]
SPI: 12345 (0x3039)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA encrypted packets (number/bytes): 125/7500
No duration limit for this SA

[Inbound ESP SAs]
SPI: 54321 (0xd431)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA decrypted packets (number/bytes): 11/660
No duration limit for this SA
Anti-replay : Disable
```



结果验证

❖ 抓包结果显示

137	359.687000	2.2.2.1	2.2.2.2	ESP	138	ESP (SPI=0x00003039)
138	359.687000	2.2.2.2	2.2.2.1	ESP	138	ESP (SPI=0x0000d431)
139	362.500000	2.2.2.1	2.2.2.2	ESP	138	ESP (SPI=0x00003039)
140	364.500000	2.2.2.1	2.2.2.2	ESP	138	ESP (SPI=0x00003039)
141	364.515000	2.2.2.2	2.2.2.1	ESP	138	ESP (SPI=0x0000d431)
142	365.515000	2.2.2.1	2.2.2.2	ESP	138	ESP (SPI=0x00003039)
143	365.515000	2.2.2.2	2.2.2.1	ESP	138	ESP (SPI=0x0000d431)

> Frame 137: 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface -, id 0

> Ethernet II, Src: HuaweiTe_a2:04:df (00:e0:fc:a2:04:df), Dst: HuaweiTe_f7:72:40 (00:e0:fc:f7:72:40)

> Internet Protocol Version 4, Src: 2.2.2.1, Dst: 2.2.2.2

▼ Encapsulating Security Payload

ESP SPI: 0x00003039 (12345)

ESP Sequence: 116

0000	00 e0 fc f7 72 40 00 e0 fc a2 04 df 08 00 45 00	r@..
0010	00 7c 00 e6 40 00 ff 32 72 63 02 02 02 01 02 02	@..2 rc.....
0020	02 02 00 00 30 39 00 00 00 74 aa 91 59 3c 00 00	...09...t.Y<..
0030	00 00 1a 59 d8 4b 00 00 00 00 00 71 b1 c5 c3 7a	...Y.K...q...z
0040	03 71 32 fd 35 ce 99 4c 23 1c 77 d1 2c 9a 1b 16	..q2.5..L#.w,...
0050	e4 8c 57 5c 83 9f 3b 47 99 ff a0 c0 44 56 df 46	..W\...;G ...DV.F
0060	5d ba d6 e9 c3 c6 3d a2 36 4d b6 f5 da 8c 75 ff	].....= 6M...u<
0070	c7 0f cb 8f a6 a5 81 82 c5 a3 92 bd e0 eb 69 3c	i<
0080	9c f7 54 67 57 ed 5f c7 6b a5	..TgW..._ k.



谢谢!